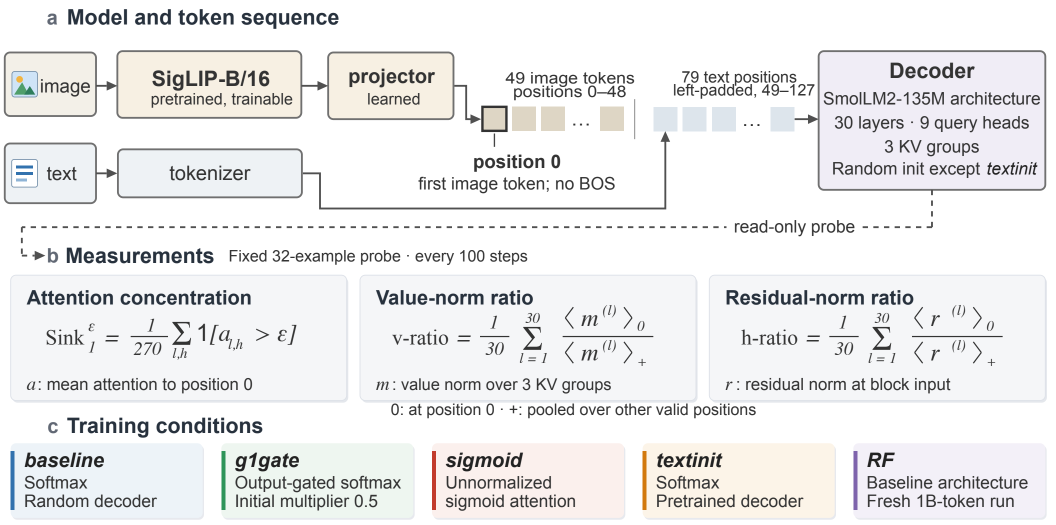
a Model and token sequence
image
SigLIP-B/16
pretrained, trainable
projector
learned
49 image tokens
positions 0–48
79 text positions
left-padded, 49–127
Decoder
SmolLM2-135M architecture
30 layers · 9 query heads
3 KV groups
Random init except textinit
text
tokenizer
position 0
first image token; no BOS
read-only probe
b Measurements Fixed 32-example probe · every 100 steps
Attention concentration
$\mathrm{Sink}_1^{\varepsilon} = \frac{1}{270}\sum_{l,h} 1[a_{l,h} > \varepsilon]$
a: mean attention to position 0
Value-norm ratio
$\text{v-ratio} = \frac{1}{30}\sum_{l=1}^{30} \frac{\langle m^{(l)} \rangle_0}{\langle m^{(l)} \rangle_+}$
m: value norm over 3 KV groups
Residual-norm ratio
$\text{h-ratio} = \frac{1}{30}\sum_{l=1}^{30} \frac{\langle r^{(l)} \rangle_0}{\langle r^{(l)} \rangle_+}$
r: residual norm at block input
0: at position 0 · +: pooled over other valid positions
c Training conditions
baseline
Softmax
Random decoder
g1gate
Output-gated softmax
Initial multiplier 0.5
sigmoid
Unnormalized
sigmoid attention
textinit
Softmax
Pretrained decoder
RF
Baseline architecture
Fresh 1B-token run